

Machine Learning Project Instructions

Project Presentation and Demonstration Guidelines

Deadline: Sunday, 28 June 2026 (No Extension)

Important message

These are applied Machine Learning projects, not simple “download data and apply a few models” tasks. Each team must understand the problem, inspect the dataset, design a suitable methodology, justify the evaluation protocol, and clearly explain what the results mean. The strongest projects will show thoughtful problem understanding, careful validation, meaningful analysis, and a smooth live demonstration.

1. Nature of the Projects

The assigned projects are based on real-world Machine Learning problems. Many of them involve issues such as distribution shift, fairness, robustness, multimodal fusion, open-set learning, long documents, noisy labels, temporal validation, or domain-specific evaluation. Therefore, the project cannot be completed by simply choosing a dataset, training a model, and reporting accuracy.

Your work should show that you understand the actual problem being solved. For example, a medical warning system may require early detection and false-alarm control; a visual question answering project may require answerability and grounding; a cybersecurity project may require careful split design to avoid leakage; and a fairness or robustness project may require subgroup-level analysis rather than only one global score.

The expected output is a complete, explainable, and demonstrable Machine Learning workflow. The methodology, assumptions, limitations, and interpretation of results are as important as the model itself.

2. First Step: Understand the Project Before Coding

Before downloading data or writing code, each team must carefully study its project title and problem statement. The title gives the general theme, but the problem statement defines what is actually required.

- Read the project title and problem statement multiple times.
- Identify the main prediction, classification, retrieval, segmentation, recommendation, or robustness question.
- Write down what the project is asking you to compare, test, or demonstrate.
- Identify any special requirement in the problem statement, such as leave-subject-out testing, drift-aware validation, open-set evaluation, answerability detection, calibration, threshold transfer, multimodal fusion, or subgroup analysis.
- Discuss in the group what would count as a good solution and what would be an incomplete or misleading solution.
- Define a realistic scope. If the dataset is too large, use a carefully selected subset, but explain why the subset is reasonable.

3. Data Review and Dataset Understanding

The dataset must be reviewed manually and technically before modelling. Do not assume that the dataset directly fits the project just because it is listed with the title.

- Download the dataset early and confirm that it is accessible. Some datasets may require registration, credentials, or special download steps.
- Open and inspect the files manually. Look at examples, labels, metadata, folder structure, file formats, and missing values.

- Check whether the dataset contains the information required by the problem statement. For example, check whether subject IDs, timestamps, domains, groups, modalities, labels, question-answer pairs, regions, or subgroup attributes are available.
- Understand the labels. Know what each class or target variable means and whether the task is binary, multiclass, multilabel, retrieval, regression, segmentation, ranking, or generation.
- Check the size of the data and decide whether full training is possible. If not, create a reproducible subset and document the sampling method.
- Check for leakage risks. Avoid random splits if the project requires split by subject, time, region, document, attack family, camera location, hospital, product, or domain.
- Create a short exploratory data analysis section in your notebook showing dataset size, label distribution, sample records/images/text/audio, missingness, and any important imbalance.

4. Group Brainstorming and Methodology Design

After understanding the problem and reviewing the data, brainstorm the methodology as a group. The methodology should connect the data, models, evaluation, and final answer to the main project question.

- Break the project into steps: data preparation, preprocessing, feature extraction, model design, validation protocol, evaluation, error analysis, and interpretation.
- Identify what the baseline should be. A simple but correct baseline is required before trying a more complex model.
- Select models that are appropriate for the data and the problem. The model choice must be justified, not random.
- Define the evaluation criterion before running experiments. The metric should match the task and the problem statement.
- For robustness, fairness, distribution shift, or safety-related projects, include subgroup/domain/shift-specific evaluation instead of reporting only a single average metric.
- For multimodal projects, compare at least one single-modality baseline with a fused or combined approach, where feasible.
- For retrieval, ranking, or recommendation tasks, use suitable retrieval metrics rather than only classification accuracy.
- For medical, security, or high-risk tasks, discuss false positives, false negatives, calibration, threshold selection, or operational consequences where relevant.
- Include error analysis. Show examples where the model succeeds and fails, and explain what those failures mean.

5. AI-Based Tool Usage

AI-based tools may be used to support the project, but the final responsibility and ownership remain fully with the team. AI tools can help with code debugging, literature search, explanation of errors, or generating initial ideas, but they do not replace your own understanding.

- Do not submit or present code that the team cannot explain.
- Verify all AI-generated code, explanations, metrics, and claims.
- Check that the code runs in your environment and produces the reported results.
- Clearly understand every major preprocessing step, model choice, metric, and output.
- During Q&A, saying that “AI generated it” is not an acceptable answer. The team must own the work.

6. Team Responsibility and Q&A

All team members must be involved and must understand the complete project, not only the part they personally implemented.

- Every team member must be able to explain the problem statement, dataset, methodology, models, evaluation, results, and limitations.

- Divide work fairly, but discuss progress regularly so that knowledge is shared across the whole team.
- Prepare for questions about why you selected a particular model, split, metric, threshold, subset, or preprocessing method.
- Be ready to explain what the results mean, not only what the numbers are.
- Be honest about limitations. A clear explanation of limitations is better than overclaiming.

7. Required Presentation and Demonstration Format

Each group must prepare one main Jupyter Notebook file (.ipynb) for the presentation and code demonstration. The notebook should be clean, executable, and suitable for presenting the project from start to finish.

Minimum required notebook sections

Section	What to Include
Title Slide	Project title, course, group number, team members, matriculation numbers, and date.
Problem Statement	Explain the project in your own words. State what question the project is trying to answer. Use bullet points instead of long text.
Dataset	Describe the dataset, source, files used, labels/targets, important variables, sample examples, size, and any subset used. Use bullet points instead of long text.
Methodology	Explain the full workflow: preprocessing, split design, feature extraction, modelling, validation, and analysis. Use bullet points instead of long text.
Selected Models	List the baseline and main models. Explain why these models are suitable for the task. Use bullet points instead of long text.
Evaluation Criterion	Define the metrics and explain why they fit the problem. Include task-specific metrics where needed. Use bullet points instead of long text.
Results	Present the main results clearly using tables and/or figures. Include comparison with the baseline. Use bullet points instead of long text.
Error Analysis and Limitations	Show important failure cases, limitations, and what could be improved. Use bullet points instead of long text.
Conclusion	Summarise the main finding and answer the original project question. Use bullet points instead of long text.
Code Demonstration	Include runnable cells for the main pipeline or a reduced demonstration that clearly shows the workflow executing.

8. Code Demonstration Requirements

- You will be required to execute the code during the presentation.
- Install all dependencies before the presentation. Do not install packages for the first time during the presentation.
- Use a clean notebook structure. Remove unnecessary failed attempts, unused code, and confusing outputs from the final presentation notebook.
- Make the code reproducible. Set random seeds where possible and clearly mention any randomness or sampling.
- Use relative paths or clearly documented paths. Avoid code that only works on one person's laptop because of hardcoded local folders.
- If the full model training is too slow, include saved outputs and a short executable demonstration that runs smoothly. However, you must still be able to explain how the full results were obtained.

- Keep a copy of important outputs, figures, trained lightweight models, and processed data needed for the demonstration.
- Test the final notebook from top to bottom before the presentation.

9. Backup and Presentation Readiness

Technical preparation is part of the project. If the team is unable to present because files, laptops, connectors, dependencies, or internet access are not ready, a separate presentation slot cannot be arranged later.

- The final version must be available online, for example on GitHub, Google Drive, OneDrive, or another shared location.
- The final version must also be available locally on all team members' laptops.
- Keep at least one backup copy of the notebook and essential files on another device or cloud folder.
- Check that the notebook opens and runs on more than one laptop.
- Bring the laptop charger.
- Make sure that you have the required USB-C connector or adapter for presentation.
- Check screen sharing or projector connection before your turn where possible.
- Do not depend on one team member only. Any member should be able to open the notebook and continue the presentation if needed.

10. Evaluation Criterion

The evaluation will focus mainly on the quality of the methodology and the quality of the Q&A. A project with a modest model but a strong, correct methodology may be evaluated better than a project with a complex model but weak problem understanding.

Criterion	What Will Be Considered
Methodology	Understanding of the problem statement; suitability of data review; correct split design; justified preprocessing; meaningful baseline; appropriate model choice; suitable evaluation metrics; careful interpretation; error analysis; limitations.
Q&A	Ability of all members to explain the full project; clarity of answers; understanding of code and results; ability to justify choices; honesty about limitations.
Demonstration readiness	Executable notebook, working dependencies, clear outputs, stable presentation setup, and proper backups. This affects the ability to assess the project properly.

11. Final Submission and Deadline

- Make sure that everything is ready by 28th June 2026
- You are not required to submit anything to me as yet.

12. Final Checklist Before Presentation

Done	Checklist Item
☐	We can explain the project title and problem statement in our own words.
☐	We manually reviewed the dataset and understand the labels, files, and limitations.
☐	We selected a validation split that matches the real project requirement and avoids leakage.
☐	We included at least one meaningful baseline.
☐	We selected evaluation metrics that match the task.

☐	We included results in a clear table or figure.
☐	We included some error analysis or failure examples.
☐	All team members understand the full workflow.
☐	The notebook runs from top to bottom or has a clearly prepared executable demonstration path.
☐	All dependencies are installed before presentation.
☐	The final files are online and also available on all team members' laptops.
☐	We have backup copies.
☐	We have the required USB-C connector or adapter.
☐	We are ready for Q&A.

A successful project is not only one that produces a model, but one that clearly shows why the model, data, validation, and results answer the problem statement.